

InverterControl Properties							
General							
Name	InverterControl	Number	16	Type	FB	Language	LAD
Numbering	Automatic						
Information							
Title		Author		Comment		Family	
Version	0.1	User-defined ID					

Name	Data type	Default value	Retain	Accessible from HMI/OPC UA/Web API	Writ-able from HMI/OPC UA/ Web API	Visible in HMI engi-neering	Setpoint	Supervi-sion	Comment
▼ Input									
ID	CONN_OUC	16#0	Non-retain	True	True	True	False		
Run	Bool	false	Non-retain	True	True	True	False		
Request	Bool	false	Non-retain	True	True	True	False		
Disconnect	Bool	false	Non-retain	True	True	True	False		
Reset	Bool	false	Non-retain	True	True	True	False		
▼ Output									
Error	Bool	false	Non-retain	True	True	True	False		
▼ InOut									
▼ Inverter	"Sunny_inver-ter"			False	False	False	False		
InvOpMod	DInt			False	False	False	False		Operating state of the inver-ter, setpoint
RemRdy	DInt			False	False	False	False		Put inverter into operation or standby
GriMng.VArMod	DInt			False	False	False	False		Utility grid system services: Selection of procedure for setpoint for reactive power
GriMng.WMod	DInt			False	False	False	False		Utility grid system services: Selection of procedure for setpoint for active power
ErrClr	DInt			False	False	False	False		Acknowledgment of the present error
ErrClr.ProErr	DInt			False	False	False	False		Acknowledging of safety-relevant errors
InvMs.TotVA	DInt			False	False	False	False		Apparent power, total
InvMs.TotW	DInt			False	False	False	False		Active power, total
InstFunc	DInt			False	False	False	False		Activation of installation functions
InvMs.TotVAr	DInt			False	False	False	False		Reactive power, total
GriMs.Hz	Real			False	False	False	False		Grid frequency
ErrLcn	DInt			False	False	False	False		Error location of the present error
ErrStt	DInt			False	False	False	False		Error status of the inverter
ErrNo	DInt			False	False	False	False		Error number of the present error
OpStt	DInt			False	False	False	False		Operating state of the inver-ter
WSpt	DInt			False	False	False	False		Active power, setpoint
VArSpt	DInt			False	False	False	False		Reactive power, setpoint
PFSpt	Real			False	False	False	False		Power factor cos (phi), set-point
WAval	Real			False	False	False	False		Currently available active power
VArAval	Real			False	False	False	False		Currently available reactive power
PwrOffReas	DInt			False	False	False	False		Current cause of inverter dis-connection
▼ ParamHold	"Skid_Parame-ters_Hold"			False	False	False	False		
VADrtPriMod	DInt			False	False	False	False		Prioritizing of control of de-rating
WGraMod	DInt			False	False	False	False		Activates the active power gradient
WGra	Real			False	False	False	False		Active power gradient
VArGraMod	DInt			False	False	False	False		Activates the reactive power gradient
VArGra	Real			False	False	False	False		Reactive power gradient
GriMng.InvVArMod	DInt			False	False	False	False		Grid management services: specification of reactive power setpoint by the inver-ter
PvGnd.OpnRemGfdi	DInt			False	False	False	False		Open the Remote GFDI
StbySfCapacMod	DInt			False	False	False	False		Activation of standby mode of the sine-wave filter capac-itor

Totally Integrated Automation Portal											
Name	Data type	Default value	Retain	Accessible from HMI/OPC UA/Web API	Writ-able from HMI/OPC UA/Web API	Visible in HMI engi-neering	Setpoint	Supervi-sion			Comment
HzNomSpt	Real			False	False	False	False				
DrtStt	DInt			False	False	False	False				
DiagRmgTm	DInt			False	False	False	False				
Rio.Din.FloatCtl.Wa	Int			False	False	False	False				
Rio.Din.FloatCtl.Err	Int			False	False	False	False				
WRtg	DInt			False	False	False	False				
Gfdi.AmpPrc	Real			False	False	False	False				
Gfdi.AmpErr	Real			False	False	False	False				
ActErrNo1	DInt			False	False	False	False				
ActErrLcn1	DInt			False	False	False	False				
ActErrNo2	DInt			False	False	False	False				
ActErrLcn2	DInt			False	False	False	False				
ActErrNo3	DInt			False	False	False	False				
ActErrLcn3	DInt			False	False	False	False				
ActErrNo4	DInt			False	False	False	False				
ActErrLcn4	DInt			False	False	False	False				
ActErrNo5	DInt			False	False	False	False				
ActErrLcn5	DInt			False	False	False	False				
ActErrNo6	DInt			False	False	False	False				
ActErrLcn6	DInt			False	False	False	False				
ActErrNo7	DInt			False	False	False	False				
ActErrLcn7	DInt			False	False	False	False				
ActErrNo8	DInt			False	False	False	False				
ActErrLcn8	DInt			False	False	False	False				
ActErrNo9	DInt			False	False	False	False				
ActErrLcn9	DInt			False	False	False	False				
ActErrNo10	DInt			False	False	False	False				
ActErrLcn10	DInt			False	False	False	False				
WSpt_Fdbk	DInt			False	False	False	False				
VarSpt_Fdbk	DInt			False	False	False	False				
PFSpt_Fdbk	Real			False	False	False	False				
▼ connectParamClient	TCON_IP_v4			False	False	False	False				
Interfaceld	HW_ANY			False	False	False	False				HW-identifier of IE-interface submodule
ID	CONN_OUC			False	False	False	False				connection reference / iden-tifier
ConnectionType	Byte			False	False	False	False				type of connection: 11=TCP/IP, 19=UDP (17=TCP/IIP)
ActiveEstablished	Bool			False	False	False	False				active/passive connection es-tablishment
▼ RemoteAddress	IP_V4			False	False	False	False				remote IP address (IPv4)
▼ ADDR	Array[1..4] of Byte			False	False	False	False				IPv4 address
ADDR[1]	Byte			False	False	False	False				IPv4 address
ADDR[2]	Byte			False	False	False	False				IPv4 address
ADDR[3]	Byte			False	False	False	False				IPv4 address
ADDR[4]	Byte			False	False	False	False				IPv4 address
RemotePort	UInt			False	False	False	False				remote UDP/TCP port num-ber
LocalPort	UInt			False	False	False	False				local UDP/TCP port number
▼ Static											
Req	Bool	false	Non-retain	True	True	True	False				
▼ clientData	Struct		Non-retain	True	True	True	False				
request	Bool	false	Non-retain	True	True	True	False				
disconnect	Bool	false	Non-retain	True	True	True	False				
done	Bool	false	Non-retain	True	True	True	False				
busy	Bool	false	Non-retain	True	True	True	False				
error	Bool	false	Non-retain	True	True	True	False				
status	Word	16#0	Non-retain	True	True	True	False				
statusSave	Word	16#0	Non-retain	True	True	True	False				
modbusMode	USInt	0	Non-retain	True	True	True	False				
modbusDataAddress	UDInt	0	Non-retain	True	True	True	False				
modbusDataLen	UInt	0	Non-retain	True	True	True	False				
▼ clientData2	Struct		Non-retain	True	True	True	False				
request	Bool	false	Non-retain	True	True	True	False				
disconnect	Bool	false	Non-retain	True	True	True	False				
done	Bool	false	Non-retain	True	True	True	False				
busy	Bool	false	Non-retain	True	True	True	False				
error	Bool	false	Non-retain	True	True	True	False				
status	Word	16#0	Non-retain	True	True	True	False				
statusSave	Word	16#0	Non-retain	True	True	True	False				
modbusMode	USInt	0	Non-retain	True	True	True	False				

Totally Integrated Automation Portal												
Name		Data type	Default value	Retain	Accessible from HMI/OPC UA/Web API	Writ-able from HMI/ OPC UA/ Web API	Visible in HMI engi-neering	Setpoint	Supervi-sion			Comment
modbusDataAddress		UDInt	0	Non-retain	True	True	True	False				
modbusDataLen		UInt	0	Non-retain	True	True	True	False				
▼ connectParamServer		TCON_IP_v4		Non-retain	True	True	True	False				
InterfaceId		HW_ANY	0	Non-retain	True	True	True	False				HW-identifier of IE-interface submodule
ID		CONN_OUC	16#0	Non-retain	True	True	True	False				connection reference / identifier
ConnectionType		Byte	16#0B	Non-retain	True	True	True	False				type of connection: 11=TCP/IP, 19=UDP (17=TCP/IP)
ActiveEstablished		Bool	false	Non-retain	True	True	True	False				active/passive connection es-tablishment
▼ RemoteAddress		IP_V4		Non-retain	True	True	True	False				remote IP address (IPv4)
▼ ADDR		Array[1..4] of Byte		Non-retain	True	True	True	False				IPv4 address
ADDR[1]		Byte	16#0	Non-retain	True	True	True	False				IPv4 address
ADDR[2]		Byte	16#0	Non-retain	True	True	True	False				IPv4 address
ADDR[3]		Byte	16#0	Non-retain	True	True	True	False				IPv4 address
ADDR[4]		Byte	16#0	Non-retain	True	True	True	False				IPv4 address
RemotePort		UInt	0	Non-retain	True	True	True	False				remote UDP/TCP port number
LocalPort		UInt	0	Non-retain	True	True	True	False				local UDP/TCP port number
▼ serverData		Struct		Non-retain	True	True	True	False				
disconnect		Bool	false	Non-retain	True	True	True	False				
ndr		Bool	false	Non-retain	True	True	True	False				
dataRead		Bool	false	Non-retain	True	True	True	False				
error		Bool	false	Non-retain	True	True	True	False				
status		Word	16#0	Non-retain	True	True	True	False				
statusSave		Word	16#0	Non-retain	True	True	True	False				
▼ holdingRegisterS1		Array[0..200] of Word		Set in IDB	True	True	True	False				
holdingRegisterS1[0]		Word	16#0	Set in IDB	True	True	True	False				
holdingRegisterS1[1]		Word	16#0	Set in IDB	True	True	True	False				
holdingRegisterS1[2]		Word	16#0	Set in IDB	True	True	True	False				
holdingRegisterS1[3]		Word	16#0	Set in IDB	True	True	True	False				
holdingRegisterS1[4]		Word	16#0	Set in IDB	True	True	True	False				
holdingRegisterS1[5]		Word	16#0	Set in IDB	True	True	True	False				
holdingRegisterS1[6]		Word	16#0	Set in IDB	True	True	True	False				
holdingRegisterS1[7]		Word	16#0	Set in IDB	True	True	True	False				
holdingRegisterS1[8]		Word	16#0	Set in IDB	True	True	True	False				
holdingRegisterS1[9]		Word	16#0	Set in IDB	True	True	True	False				
holdingRegisterS1[10]		Word	16#0	Set in IDB	True	True	True	False				
holdingRegisterS1[11]		Word	16#0	Set in IDB	True	True	True	False				
holdingRegisterS1[12]		Word	16#0	Set in IDB	True	True	True	False				
holdingRegisterS1[13]		Word	16#0	Set in IDB	True	True	True	False				
holdingRegisterS1[14]		Word	16#0	Set in IDB	True	True	True	False				
holdingRegisterS1[15]		Word	16#0	Set in IDB	True	True	True	False				
holdingRegisterS1[16]		Word	16#0	Set in IDB	True	True	True	False				
holdingRegisterS1[17]		Word	16#0	Set in IDB	True	True	True	False				
holdingRegisterS1[18]		Word	16#0	Set in IDB	True	True	True	False				
holdingRegisterS1[19]		Word	16#0	Set in IDB	True	True	True	False				
holdingRegisterS1[20]		Word	16#0	Set in IDB	True	True	True	False				
holdingRegisterS1[21]		Word	16#0	Set in IDB	True	True	True	False				
holdingRegisterS1[22]		Word	16#0	Set in IDB	True	True	True	False				
holdingRegisterS1[23]		Word	16#0	Set in IDB	True	True	True	False				
holdingRegisterS1[24]		Word	16#0	Set in IDB	True	True	True	False				
holdingRegisterS1[25]		Word	16#0	Set in IDB	True	True	True	False				
holdingRegisterS1[26]		Word	16#0	Set in IDB	True	True	True	False				
holdingRegisterS1[27]		Word	16#0	Set in IDB	True	True	True	False				
holdingRegisterS1[28]		Word	16#0	Set in IDB	True	True	True	False				
holdingRegisterS1[29]		Word	16#0	Set in IDB	True	True	True	False				
holdingRegisterS1[30]		Word	16#0	Set in IDB	True	True	True	False				
holdingRegisterS1[31]		Word	16#0	Set in IDB	True	True	True	False				
holdingRegisterS1[32]		Word	16#0	Set in IDB	True	True	True	False				
holdingRegisterS1[33]		Word	16#0	Set in IDB	True	True	True	False				
holdingRegisterS1[34]		Word	16#0	Set in IDB	True	True	True	False				
holdingRegisterS1[35]		Word	16#0	Set in IDB	True	True	True	False				
holdingRegisterS1[36]		Word	16#0	Set in IDB	True	True	True	False				
holdingRegisterS1[37]		Word	16#0	Set in IDB	True	True	True	False				
holdingRegisterS1[38]		Word	16#0	Set in IDB	True	True	True	False				
holdingRegisterS1[39]		Word	16#0	Set in IDB	True	True	True	False				
holdingRegisterS1[40]		Word	16#0	Set in IDB	True	True	True	False				
holdingRegisterS1[41]		Word	16#0	Set in IDB	True	True	True	False				
holdingRegisterS1[42]		Word	16#0	Set in IDB	True	True	True	False				

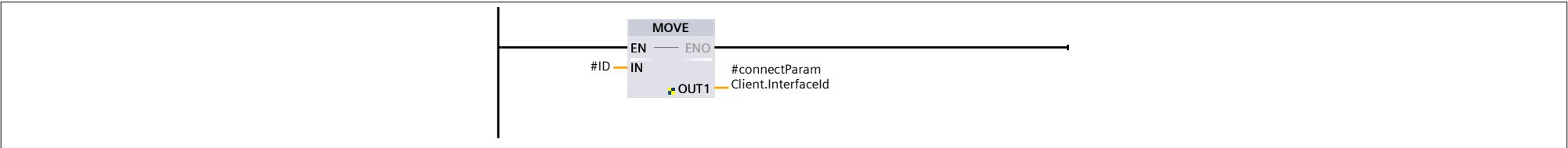
Totally Integrated Automation Portal													
Name		Data type	Default value	Retain	Accessible from HMI/OPC UA/Web API	Writ-able from HMI/ OPC UA/ Web API	Visible in HMI engi-neering	Setpoint	Supervi-sion			Comment	
Dints1[34]	DInt	0	Non-retain	True	True	True	False						
Dints1[35]	DInt	0	Non-retain	True	True	True	False						
Dints1[36]	DInt	0	Non-retain	True	True	True	False						
Dints1[37]	DInt	0	Non-retain	True	True	True	False						
Dints1[38]	DInt	0	Non-retain	True	True	True	False						
Dints1[39]	DInt	0	Non-retain	True	True	True	False						
Dints1[40]	DInt	0	Non-retain	True	True	True	False						
Dints1[41]	DInt	0	Non-retain	True	True	True	False						
Dints1[42]	DInt	0	Non-retain	True	True	True	False						
Dints1[43]	DInt	0	Non-retain	True	True	True	False						
Dints1[44]	DInt	0	Non-retain	True	True	True	False						
Dints1[45]	DInt	0	Non-retain	True	True	True	False						
Dints1[46]	DInt	0	Non-retain	True	True	True	False						
Dints1[47]	DInt	0	Non-retain	True	True	True	False						
Dints1[48]	DInt	0	Non-retain	True	True	True	False						
Dints1[49]	DInt	0	Non-retain	True	True	True	False						
Dints1[50]	DInt	0	Non-retain	True	True	True	False						
Dints1[51]	DInt	0	Non-retain	True	True	True	False						
Dints1[52]	DInt	0	Non-retain	True	True	True	False						
Dints1[53]	DInt	0	Non-retain	True	True	True	False						
Dints1[54]	DInt	0	Non-retain	True	True	True	False						
Dints1[55]	DInt	0	Non-retain	True	True	True	False						
Dints1[56]	DInt	0	Non-retain	True	True	True	False						
Dints1[57]	DInt	0	Non-retain	True	True	True	False						
Dints1[58]	DInt	0	Non-retain	True	True	True	False						
Dints1[59]	DInt	0	Non-retain	True	True	True	False						
Dints1[60]	DInt	0	Non-retain	True	True	True	False						
Dints1[61]	DInt	0	Non-retain	True	True	True	False						
Dints1[62]	DInt	0	Non-retain	True	True	True	False						
Dints1[63]	DInt	0	Non-retain	True	True	True	False						
Dints1[64]	DInt	0	Non-retain	True	True	True	False						
Dints1[65]	DInt	0	Non-retain	True	True	True	False						
Dints1[66]	DInt	0	Non-retain	True	True	True	False						
Dints1[67]	DInt	0	Non-retain	True	True	True	False						
Dints1[68]	DInt	0	Non-retain	True	True	True	False						
Dints1[69]	DInt	0	Non-retain	True	True	True	False						
Dints1[70]	DInt	0	Non-retain	True	True	True	False						
Dints1[71]	DInt	0	Non-retain	True	True	True	False						
Dints1[72]	DInt	0	Non-retain	True	True	True	False						
Dints1[73]	DInt	0	Non-retain	True	True	True	False						
Dints1[74]	DInt	0	Non-retain	True	True	True	False						
Dints1[75]	DInt	0	Non-retain	True	True	True	False						
Dints1[76]	DInt	0	Non-retain	True	True	True	False						
Dints1[77]	DInt	0	Non-retain	True	True	True	False						
Dints1[78]	DInt	0	Non-retain	True	True	True	False						
Dints1[79]	DInt	0	Non-retain	True	True	True	False						
Dints1[80]	DInt	0	Non-retain	True	True	True	False						
Dints1[81]	DInt	0	Non-retain	True	True	True	False						
Dints1[82]	DInt	0	Non-retain	True	True	True	False						
Dints1[83]	DInt	0	Non-retain	True	True	True	False						
Dints1[84]	DInt	0	Non-retain	True	True	True	False						
Dints1[85]	DInt	0	Non-retain	True	True	True	False						
Dints1[86]	DInt	0	Non-retain	True	True	True	False						
Dints1[87]	DInt	0	Non-retain	True	True	True	False						
Dints1[88]	DInt	0	Non-retain	True	True	True	False						
Dints1[89]	DInt	0	Non-retain	True	True	True	False						
Dints1[90]	DInt	0	Non-retain	True	True	True	False						
Dints1[91]	DInt	0	Non-retain	True	True	True	False						
Dints1[92]	DInt	0	Non-retain	True	True	True	False						
Dints1[93]	DInt	0	Non-retain	True	True	True	False						
Dints1[94]	DInt	0	Non-retain	True	True	True	False						
Dints1[95]	DInt	0	Non-retain	True	True	True	False						
Dints1[96]	DInt	0	Non-retain	True	True	True	False						
Dints1[97]	DInt	0	Non-retain	True	True	True	False						
Dints1[98]	DInt	0	Non-retain	True	True	True	False						
Dints1[99]	DInt	0	Non-retain	True	True	True	False						
Dints1[100]	DInt	0	Non-retain	True	True	True	False						
▼ Dints2	Array[0..100] of DInt		Non-retain	True	True	True	False						
Dints2[0]	DInt	0	Non-retain	True	True	True	False						
Dints2[1]	DInt	0	Non-retain	True	True	True	False						
Dints2[2]	DInt	0	Non-retain	True	True	True	False						
Dints2[3]	DInt	0	Non-retain	True	True	True	False						

Totally Integrated Automation Portal													
Name		Data type	Default value	Retain	Accessible from HMI/OPC UA/Web API	Writ-able from HMI/ OPC UA/ Web API	Visible in HMI engi-neering	Setpoint	Supervi-sion			Comment	
RD_MB_DATA_PTR		Variant			False	False	False	False				Reference to the memory area to store the data that is to be read from the Modbus server	
WR_MB_DATA_PTR		Variant			False	False	False	False				Reference to the memory area to retrieve the data that is to be written to the Mod-bus server	
▼ Static													
▼ TCON		TCON			True	True	True	True				Local instance of the instruc-tion TCON	
▼ Input													
REQ		Bool	false	Non-retain	True	True	True	False				Function to be executed on rising edge	
ID		CONN_OUC	16#0	Non-retain	True	True	True	False				Connection identifier	
▼ Output													
DONE		Bool	false	Non-retain	True	True	True	False				New data received	
BUSY		Bool	false	Non-retain	True	True	True	False				Function busy	
ERROR		Bool	false	Non-retain	True	True	True	False				Error detected	
STATUS		Word	W#16#7000	Non-retain	True	True	True	False				Function result/error mes-sage	
▼ InOut													
CONNECT		Variant			False	False	False	False				Connection description	
Static													
▼ TDISCON		TDISCON			True	True	True	True				Local instance of the instruc-tion TDISCON	
▼ Input													
REQ		Bool	False	Non-retain	True	True	True	False				Function to be executed on rising edge	
ID		CONN_OUC	W#16#0	Non-retain	True	True	True	False				Connection identifier	
▼ Output													
DONE		Bool	False	Non-retain	True	True	True	False				Function performed	
BUSY		Bool	False	Non-retain	True	True	True	False				Function busy	
ERROR		Bool	False	Non-retain	True	True	True	False				Error detected	
STATUS		Word	W#16#7000	Non-retain	True	True	True	False				Function result/error mes-sage	
InOut													
Static													
▼ TSEND		TSEND			True	True	True	True				Local instance of the instruc-tion TSEND	
▼ Input													
REQ		Bool	false	Non-retain	True	True	True	False				Function to be executed on rising edge	
ID		CONN_OUC	16#0	Non-retain	True	True	True	False				Connection identifier	
LEN		UDInt	0	Non-retain	True	True	True	False				Data length to send	
▼ Output													
DONE		Bool	false	Non-retain	True	True	True	False				Send performed	
BUSY		Bool	false	Non-retain	True	True	True	False				Function busy	
ERROR		Bool	false	Non-retain	True	True	True	False				Error detected	
STATUS		Word	W#16#7000	Non-retain	True	True	True	False				Function result/error mes-sage	
▼ InOut													
DATA		Variant			False	False	False	False				Pointer on data area to send	
ADDR		Variant			False	False	False	False				Pointer on address of receiv-er	
Static													
▼ TRECEIVE		TRCV			True	True	True	True				Local instance of the instruc-tion TRCV	
▼ Input													
EN_R		Bool	false	Non-retain	True	True	True	False				EN_R=1: function enabled	
ID		CONN_OUC	16#0	Non-retain	True	True	True	False				Connection identifier	
LEN		UDInt	0	Non-retain	True	True	True	False				Data length to receive	
ADHOC		Bool	false	Non-retain	True	True	True	False				Request adhoc mode	
▼ Output													
NDR		Bool	false	Non-retain	True	True	True	False				New data received	
BUSY		Bool	false	Non-retain	True	True	True	False				Function busy	
ERROR		Bool	false	Non-retain	True	True	True	False				Error detected	
STATUS		Word	W#16#7000	Non-retain	True	True	True	False				Function result/error mes-sage	
RCVD_LEN		UDInt	0	Non-retain	True	True	True	False				Length of received data	
▼ InOut													
DATA		Variant			False	False	False	False				Buffer for received data	
ADDR		Variant			False	False	False	False				Address of sender	
Static													

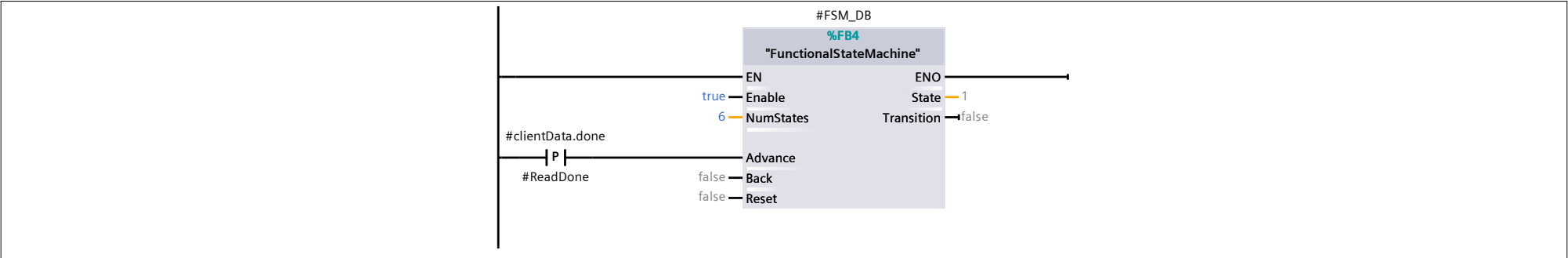
Totally Integrated Automation Portal												
Name		Data type	Default value	Retain	Accessible from HMI/OPC UA/Web API	Writ-able from HMI/ OPC UA/ Web API	Visible in HMI engi-neering	Setpoint	Supervi-sion			Comment
▼ TRESET		T_RESET			True	True	True	True				Local instance of the instruc-tion T_RESET
▼ Input												
REQ		Bool	false	Non-retain	True	True	True	False				Function to be executed on rising edge
ID		CONN_OUC	16#0	Non-retain	True	True	True	False				Connection identifier
▼ Output												
DONE		Bool	false	Non-retain	True	True	True	False				Function performed
BUSY		Bool	false	Non-retain	True	True	True	False				Function busy
ERROR		Bool	false	Non-retain	True	True	True	False				Error detected
STATUS		Word	16#0	Non-retain	True	True	True	False				Function result/error mes-sage
InOut												
Static												
▼ TDIAG		T_DIAG			True	True	True	True				Local instance of the instruc-tion T_DIAG
▼ Input												
REQ		Bool	false	Non-retain	True	True	True	False				Function to be executed on rising edge
ID		CONN_OUC	16#0	Non-retain	True	True	True	False				Connection identifier
▼ Output												
DONE		Bool	false	Non-retain	True	True	True	False				Function completed
BUSY		Bool	false	Non-retain	True	True	True	False				Function busy
ERROR		Bool	false	Non-retain	True	True	True	False				Error detected
STATUS		Word	16#0	Non-retain	True	True	True	False				Function result/error mes-sage
▼ InOut												
RESULT		Variant			False	False	False	False				Diagnostics information
Static												
▼ TDIAG_Status		TDiag_Status		Non-retain	True	True	True	True				TDiag_Status structure
Interfaceld		HW_ANY	0	Non-retain	True	True	True	False				HW-identifier of IE-interface submodule
ID		CONN_OUC	16#0	Non-retain	True	True	True	False				connection reference / iden-tifier of monitored connec-tion
ConnectionType		Byte	16#00	Non-retain	True	True	True	False				type of monitored connetion
ActiveEstablished		Bool	false	Non-retain	True	True	True	False				active/passive connection es-tablishment
State		Byte	16#00	Non-retain	True	True	True	False				state of monitored connec-tion
Kind		Byte	16#00	Non-retain	True	True	True	False				kind of monitored connec-tion
SentBytes		UDInt	16#00	Non-retain	True	True	True	False				bytes sent via monitored connection
ReceivedBytes		UDInt	16#00	Non-retain	True	True	True	False				bytes received on monitored connection
Blocked_Proc_Timeout		Real	3.0	Non-retain	True	True	True	False				Time to wait upon a blocked Modbus Client instance be-fore setting inactive
Rcv_Timeout		Real	2.0	Non-retain	True	True	True	False				Amount of time that the cli-ent waits for the server to respond to a request
MB_Unit_ID		Byte	16#FF	Non-retain	True	True	True	False				The Modbus slave address
MB_Transaction_ID		Word	1	Non-retain	True	True	True	False				Number of the current trans-action
RD_MB_DA-TA_PTR_Length		UDInt	0	Non-retain	True	True	True	False				
WR_MB_DA-TA_PTR_Length		UDInt	0	Non-retain	True	True	True	False				
MB_State		Word	16#0	Non-retain	True	False	True	False				Internal state of the Modbus client operation
SAVED_MB_DATA_ADDR		UDInt	0	Non-retain	False	False	False	False				For internal use only
SAVED_DATA_LEN		UInt	0	Non-retain	False	False	False	False				For internal use only
SAVED_MB_MODE		USInt	0	Non-retain	False	False	False	False				For internal use only
Connection_ID		Word	16#0	Non-retain	True	True	True	False				For internal use only
Retries		Word	3	Non-retain	True	True	True	False				Number of retries that the client will attempt before re-turning a no response error
▼ TEMP_DATA		Array[1..263] of Byte		Set in IDB	True	False	True	False				For internal use only
TEMP_DATA[1]		Byte	16#0	Set in IDB	True	False	True	False				For internal use only
TEMP_DATA[2]		Byte	16#0	Set in IDB	True	False	True	False				For internal use only
TEMP_DATA[3]		Byte	16#0	Set in IDB	True	False	True	False				For internal use only
TEMP_DATA[4]		Byte	16#0	Set in IDB	True	False	True	False				For internal use only
TEMP_DATA[5]		Byte	16#0	Set in IDB	True	False	True	False				For internal use only
TEMP_DATA[6]		Byte	16#0	Set in IDB	True	False	True	False				For internal use only
TEMP_DATA[7]		Byte	16#0	Set in IDB	True	False	True	False				For internal use only

Totally Integrated Automation Portal											
Name	Data type	Default value	Retain	Accessible from HMI/OPC UA/Web API	Writ-able from HMI/OPC UA/ Web API	Visible in HMI engi-neering	Setpoint	Supervi-sion			Comment
ERROR	Bool	false	Non-retain	True	True	True	False				FALSE: Job not yet started or still in progress; TRUE: Job completed with an error. See STATUS for more infor-mation
STATUS	Word	16#0	Non-retain	True	True	True	False				Status of the instruction.
▼ InOut											
ID	CONN_OUC	16#0	Non-retain	True	True	True	False				Reference to the Connection ID
OPTION	Variant			False	False	False	False				Reference to the option de-scription
Static											
▼ ModbusClientInfo	Struct		Non-retain	True	True	True	False				
DoneTrig	Bool	false	Non-retain	True	True	True	False				
Read1Trig	Bool	false	Non-retain	True	True	True	False				
Read2Trig	Bool	false	Non-retain	True	True	True	False				
Read3Trig	Bool	false	Non-retain	True	True	True	False				
Read4Trig	Bool	false	Non-retain	True	True	True	False				
ResetState	Bool	false	Non-retain	True	True	True	False				
RequestID	Bool	false	Non-retain	True	True	True	False				
Disconnect_intern	Bool	false	Non-retain	True	True	True	False				
▼ FSM_DB	"FunctionalSta-teMachine"			True	True	True	True				
▼ Input											
Enable	Bool	false	Non-retain	True	True	True	False				
NumStates	UInt	1	Non-retain	True	True	True	False				
Advance	Bool	false	Non-retain	True	True	True	False				
Back	Bool	false	Non-retain	True	True	True	False				
Reset	Bool	false	Non-retain	True	True	True	False				
▼ Output											
State	UInt	1	Non-retain	True	True	True	False				
Transition	Bool	false	Non-retain	True	True	True	False				
InOut											
▼ Static											
adv_prev	Bool	false	Non-retain	True	True	True	False				
back_prev	Bool	false	Non-retain	True	True	True	False				
rst_prev	Bool	false	Non-retain	True	True	True	False				
adv_rise	Bool	false	Non-retain	True	True	True	False				
back_rise	Bool	false	Non-retain	True	True	True	False				
rst_rise	Bool	false	Non-retain	True	True	True	False				
nextState	UInt	0	Non-retain	True	True	True	False				
ReadDone	Bool	false	Non-retain	True	True	True	True				
▼ Temp											
index	Int										
i	Int										
Constant											

Network 1: Set HW interface



Network 2: FSM



Network 3: Request data

Totally Integrated Automation Portal

#Read1

#Read2

#Read3

#Read4

#clientData.status

==

Word

W#16#7004

#clientData.done

#Req

Network 4: Step 1 SETTINGS

0001 IF (#FSM_DB.State = 1) THEN
0002 #clientData.modbusMode := 103;
0003 #clientData.modbusDataAddress := 0;
0004 #clientData.modbusDataLen := 104;
0005 #Client_DB.MB_Unit_ID := 3;
0006
0007 END_IF;

Network 5: Step 2 SETTINGS

0001 IF (#FSM_DB.State = 2) THEN
0002 #clientData.modbusMode := 104;
0003 #clientData.modbusDataAddress := 10;
0004 #clientData.modbusDataLen := 106;
0005 #Client_DB.MB_Unit_ID := 3;
0006
0007 END_IF;

Network 6: Step 3 SETTINGS

0001 IF (#FSM_DB.State = 3) THEN
0002 #clientData.modbusMode := 104;
0003 #clientData.modbusDataAddress := 116;
0004 #clientData.modbusDataLen := 112;
0005 #Client_DB.MB_Unit_ID := 3;
0006 END_IF;

Network 7: // Network: Step 4 SETTINGS — Write control group (reg 0-9)

0001 IF (#FSM_DB.State = 4) THEN
0002 #clientData.modbusMode := 1; // FC16 Write Multiple Regs
0003 #clientData.modbusDataAddress := 0;
0004 #clientData.modbusDataLen := 10;
0005 #Client_DB.MB_Unit_ID := 3;
0006 END_IF;

Network 8: Network: Step 5 SETTINGS — Write WSpt (reg 108-109)

0001 IF (#FSM_DB.State = 5) THEN
0002 #clientData.modbusMode := 1;
0003 #clientData.modbusDataAddress := 108;
0004 #clientData.modbusDataLen := 2;
0005 #Client_DB.MB_Unit_ID := 3;
0006 END_IF;

Network 9: // Network: Step 6 SETTINGS — Write VARspt + PFSpt (reg 112-115)

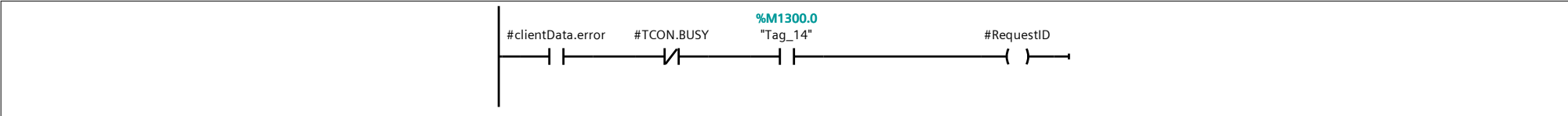
0001 IF (#FSM_DB.State = 6) THEN
0002 #clientData.modbusMode := 1;
0003 #clientData.modbusDataAddress := 112;
0004 #clientData.modbusDataLen := 4;
0005 #Client_DB.MB_Unit_ID := 3;
0006 END_IF;

Network 10:

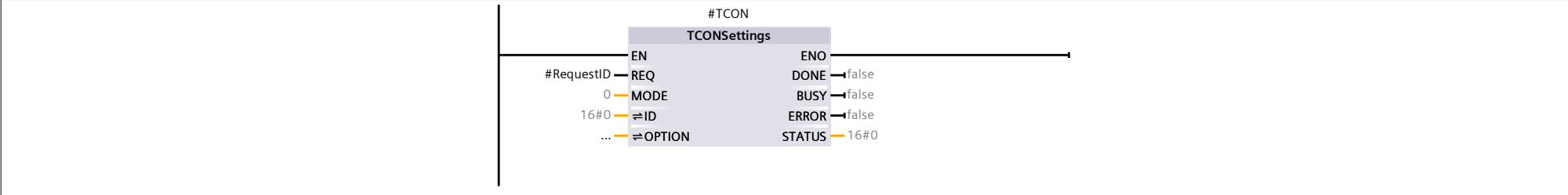
Network 11: Network: Pack RegBuf for State 4 (SCL inline or LAD EN block)

```
0001 // Network: Pack for State 4
0002 IF (#FSM_DB.State = 4) AND (#FSM_DB.Transition) THEN
0003     "FC_Pack_Write_Regs"(WriteStep := 1,
0004                           Inv := #Inverter,
0005                           RegBuf := #holdingRegisterMod);
0006 END_IF;
0007
0008 // Network: Pack for State 5
0009 IF (#FSM_DB.State = 5) AND (#FSM_DB.Transition) THEN
0010     "FC_Pack_Write_Regs"(WriteStep := 2,
0011                           Inv := #Inverter,
0012                           RegBuf := #holdingRegisterMod);
0013     #Inverter.ErrClr := 0; // State 4 just completed – safe to clear the one-shot
0014 END_IF;
0015
0016 // Network: Pack for State 6
0017 IF (#FSM_DB.State = 6) AND (#FSM_DB.Transition) THEN
0018     "FC_Pack_Write_Regs"(WriteStep := 3,
0019                           Inv := #Inverter,
0020                           RegBuf := #holdingRegisterMod);
0021 END_IF;
```

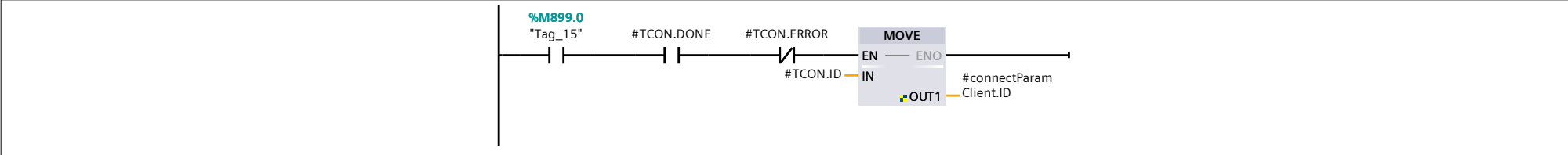
Network 12:



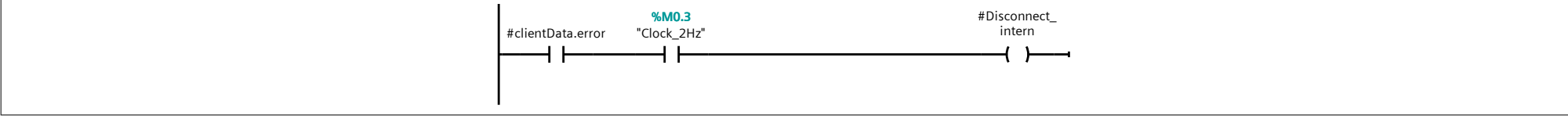
Network 13:



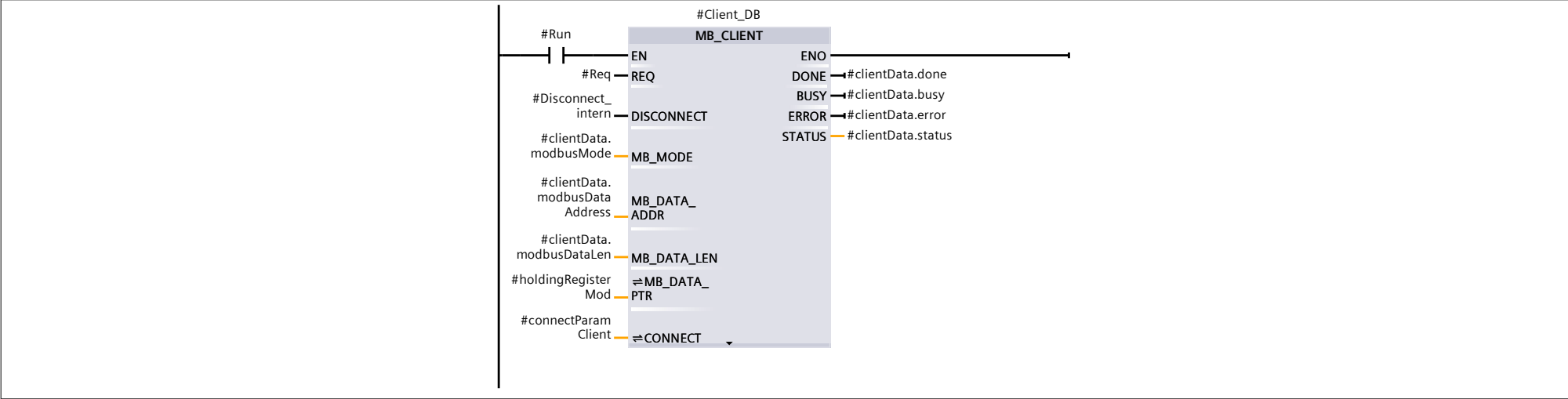
Network 14:



Network 15:



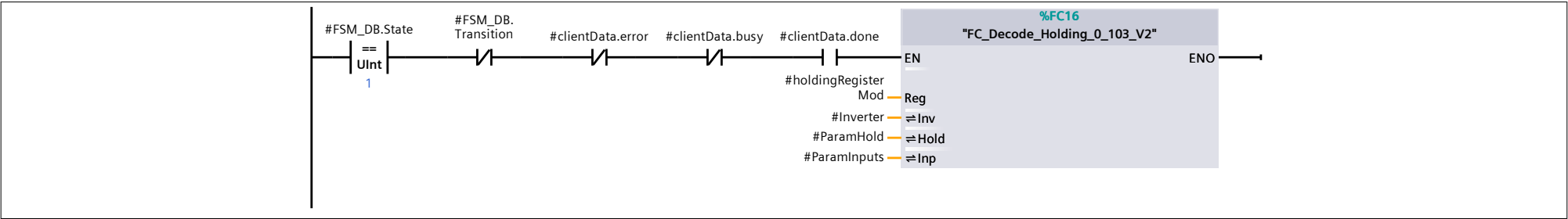
Network 16: ReadModbus



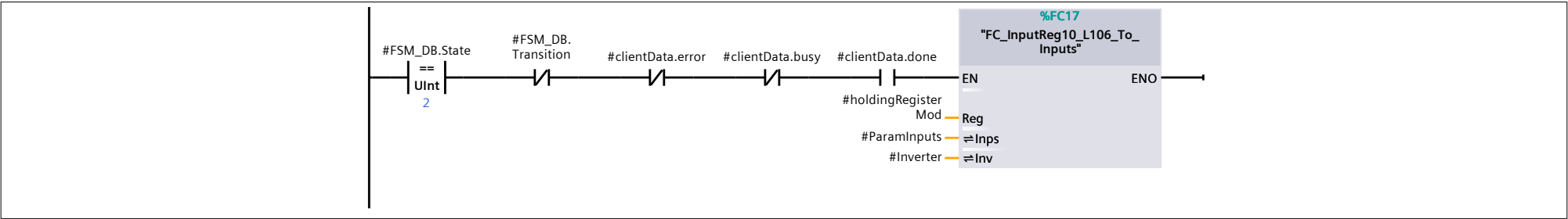
Network 17:



Network 18: STEP 1 Map Holdings



Network 19: STEP 2 MAP INPUTS



Network 20: STEP 3 MAP INPUTS

